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A Rust-Native Embeddable Reactive Numerical Language for Heterogeneous Computing in Robotics

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Motivation

Rust’s memory safety and performance make it a strong foundation for robotics. Still, numerical kernels can require substantial code for SIMD layouts and GPU execution. Backend-specific implementations can be harder to verify against the equations and maintain.

This case study presents Mech’s numerical features. Mech also has first-class state machines and pattern matching. [Scan the QR code.](#)



Embeddable

Mech runs standalone or embeds in Rust, from a single numerical function to a reactive subsystem. Rust hosts can compile Mech source or load a dylib. Browser applications can embed it through WebAssembly.

JIT compilation from Rust

```
use mech::kernel::{Backend, Kernel};
let source = include_str!("ekf.mec");
let kernel = Kernel::from_source(source)
    .input("bearing", [-0.55; 4])
    .export("state")
    .compile(Backend::Jit)?;
let mut ekf = kernel.start()?;
ekf.turn([("bearing", [-0.54; 4])]);
let state = ekf.state("state");
```

Compile once, then submit successive measurement updates. Each checked turn retains the last accepted state.

Checked dylibs[†]

Artifact	Bytes	M/s ± MAD	Peak RSS (MiB)
Mech scalar	33,544	14.671 ± 0.008	2.69 ± 0.00
Rust scalar	50,016	21.121 ± 0.005	2.69 ± 0.00
Mech SIMD-4	33,864	34.863 ± 0.010	2.69 ± 0.00

Load an AOT library

```
let kernel = unsafe { Kernel::load_bundle("ekf.bundle")? };
let mut ekf = kernel.start()?;
ekf.turn([("bearing", [-0.54; 4])]);
```

The bundle stores the dylib and metadata for named inputs, exported state and initial values.

Numerical

Mech offers a concrete syntax that expresses numerical algorithms as typed matrix equations. The bearing-only EKF below expresses prediction and correction at the matrix level.

SIMD Rust uses almost five times as many normalized source characters here, mainly for SIMD and batch support. Reusing the Mech source across backends reduces the specialized implementations to maintain.

Mech compiles a typed dataflow artifact. Its ordinary backend uses nalgebra. Other backends specialize the graph without backend-specific source annotations.

Extended Kalman Filter

```
+> math/*
Δt := 0.1
m := [140 12]'
Q := [0.01 0      -- +> import library
      0      0.0025] -- ~ mutable
R := 0.25          -- := define
I := [1 0 0      -- = assign
      0 1 0      -- ** matrix multiply
      0 0 1]      -- ' transpose
~μ := [55 25 0.4]' -- * broadcast multiply
~Σ := [100 0 0     -- x! integrity constraint
      0 100 0
      0 0 0.15]
```

Time Update

```
θ := μ[3]
d := v * Δt
G := [1 0 (0 - d * sin(θ))
      0 1 d * cos(θ)
      0 0 1]
V := [cos(θ) * Δt 0
      sin(θ) * Δt 0
      0           Δt]
```

```
μ̄ := μ + [d * cos(θ)
          d * sin(θ)
          ω * Δt]
```

```
Σ̄ := G ** Σ ** G' + V ** Q ** V'
```

Measurement Update

```
δ := m - μ̄[1..=2]
δx := δ[1]
δy := δ[2]
q := δx ^ 2 + δy ^ 2

ẑ := atan2(δy, δx) - μ̄[3]
r := z - ẑ
v := atan2(sin(r), cos(r))

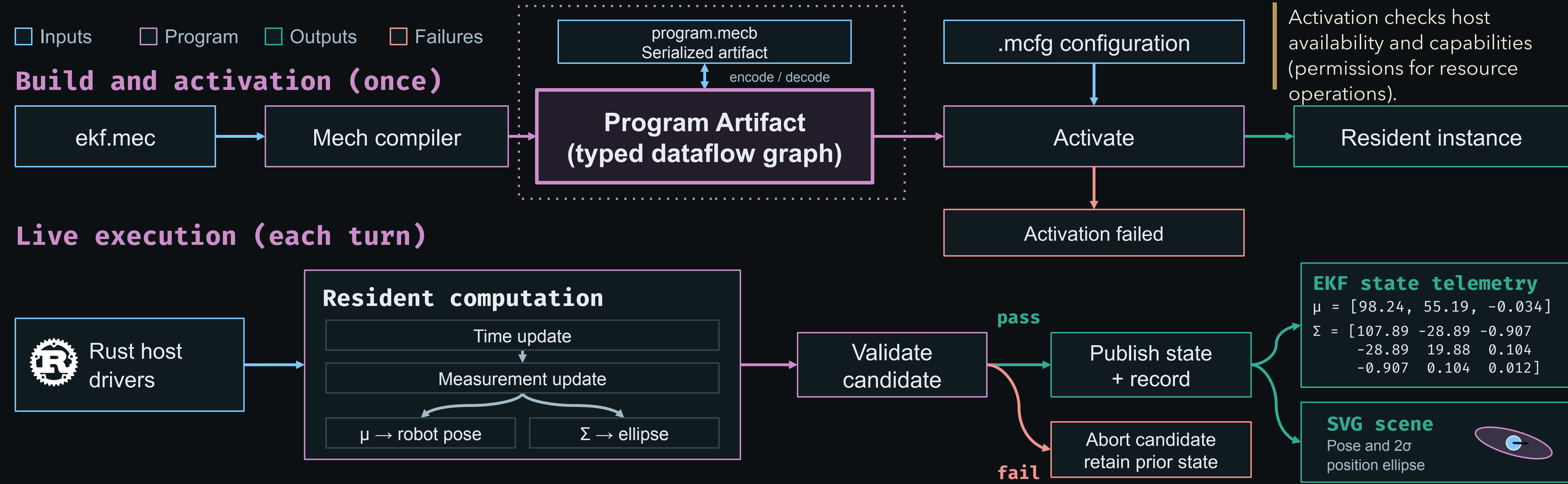
H := [δy / q, (0 - δx) / q, -1]
S := H ** Σ̄ ** H' + R
K := (Σ̄ ** H') / S[1]
A := I - K ** H

μ+ := μ̄ + K * v
Σ+ := A ** Σ̄ ** A' + (K ** K') * R

μ = μ+
Σ = Σ+
variance! := Σ+[1,1] > 0
```

Reactive

Robots operate feedback loops over continuous data streams. Rust host drivers supply data directly or through ROS. Each turn checks candidate state against integrity constraints. Rejection notifies the caller and retains prior state.

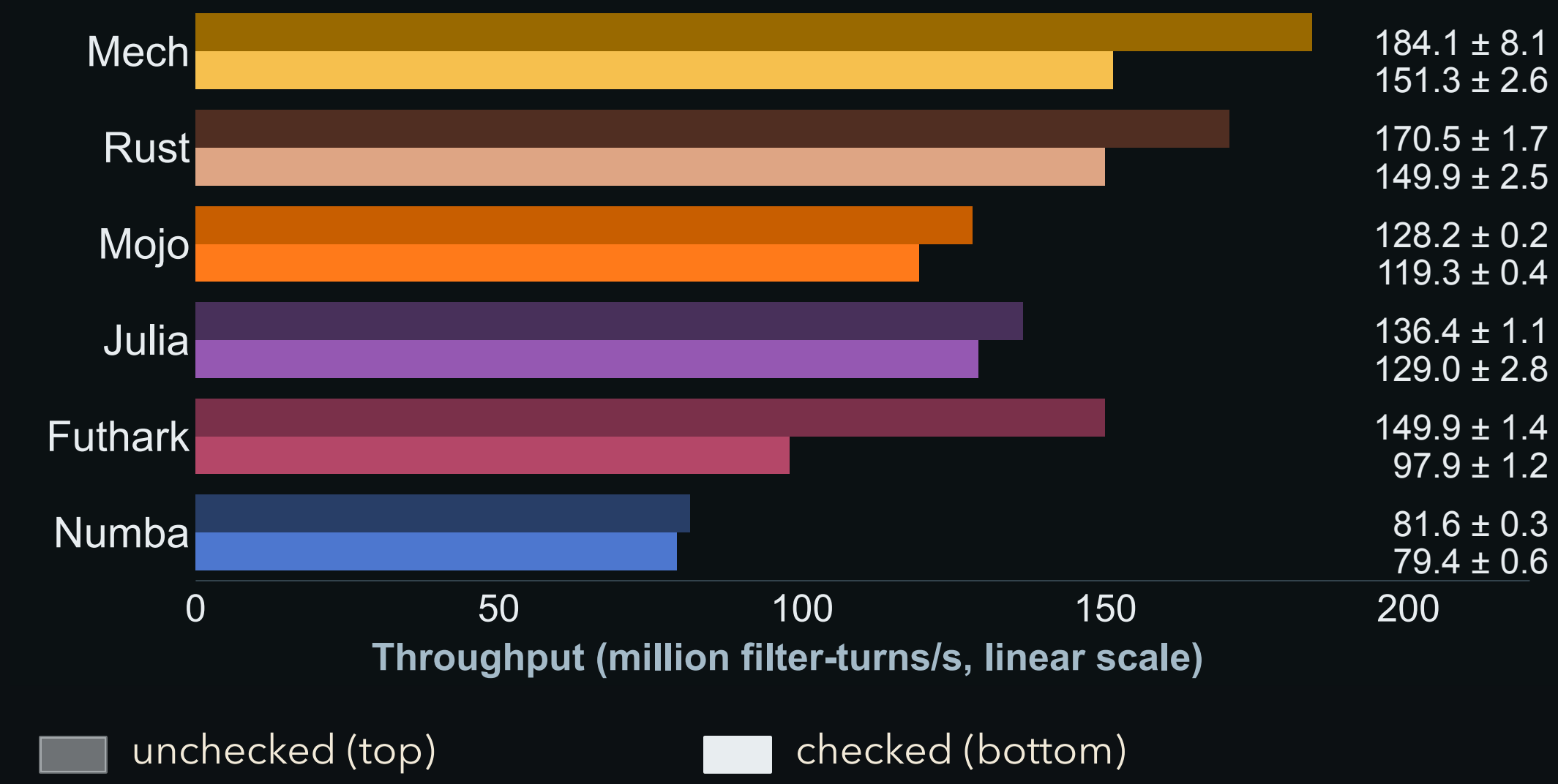


Heterogeneous

Mech’s typed graph supports CPU, GPU and WebAssembly/WGPU; the right chart runs one f32 EKF implementation of the displayed equations through five backend configurations without source changes, checked and unchecked. Integrity checks add a runtime cost. Disabling them improves throughput without removing the reactive runtime.

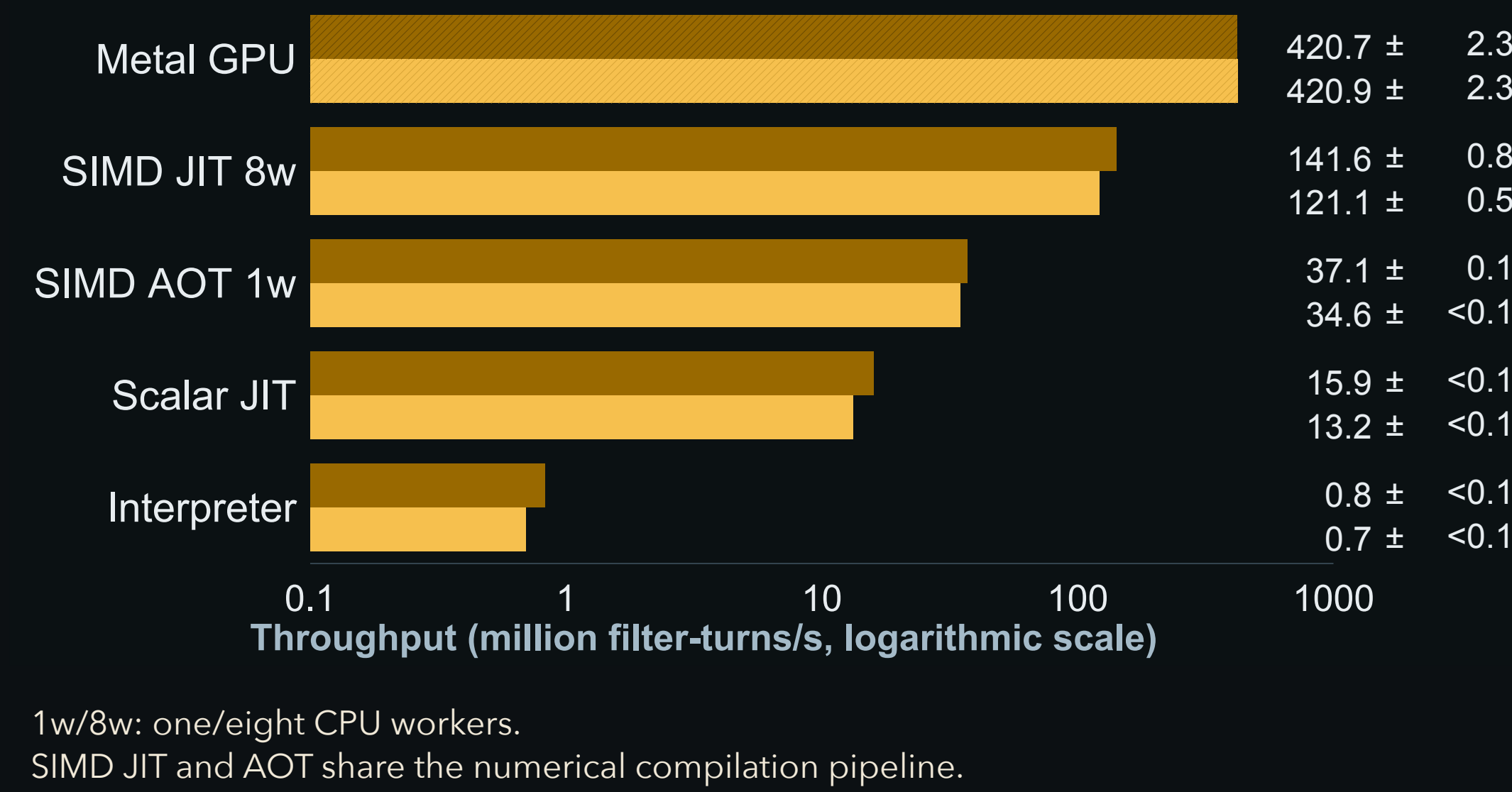
Mech and Rust show similar throughput[‡]

8 workers, fused execution. Median ± MAD. Mech: Cranelift. Rust: LLVM.



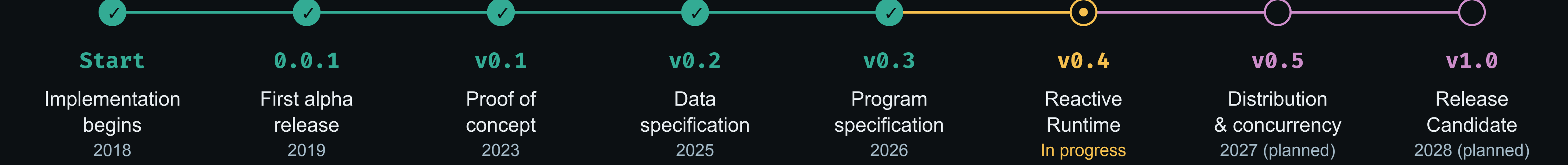
Backend choice changes EKF throughput

Per-turn publication, median ± MAD



Status and Future Work

Mech v0.4-beta is a prerelease with implemented features that need further validation and hardening. Over the next year, we plan to expand limited deployments on research robots, gather feedback, and implement additional hosts and language features toward a v1.0 release in 2028.



Experience this poster as a live reactive program: CPU/WGPU in WebAssembly. Contributions and bug reports are welcome on GitHub: [mech-lang/mech](#).



‡ Benchmark setup. Apple M1: 4P + 4E CPU, 8-core GPU, 8 GB, macOS 15.6.1. Charts: f32, 500,000 filters × 40 turns, n=10. Median ± MAD. Rust 1.96.0-nightly ec818fda3/LLVM 22.1.0. Cranelift 0.131.3. Julia 1.12.6. Mojo 1.1.0/MAX 26.6.0. Futhark 0.27.1/SPC 1.31.0/LLVM 22.1.8. Python 3.12.14/NumPy 2.2.6/Numba 0.61.2. Clang 17.0.0. Mech v0.4 development (package 0.3.6). Backend pairs f4b69052c, CPU eedc1c75b, dylibs dab98af1d. iros-workshop-benchmarks, benchmarks/iros-2026. Dylibs: minimal native-ABI hosts, 1 worker, 10,000 filters × 200 turns, n=7. Whole-process peak RSS. Timing excludes compilation, loading and warmup.

